

Examples: State the derivatives of each function

a) $f(x) = x^{100}$

b) $f(x) = \pi$

c) $f(x) = -7.39$

$f'(x) = 100x^{99}$

$f'(x) = 0$

$f'(x) = 0$

d) $y = 9x^5$

e) $f(x) = (2x)^4$

f) $y = (x+1)^2$

$y' = 45x^4$

$f(x) = 16x^4$

$y = x^2 + 2x + 1$

$f'(x) = 64x^3$

$y' = 2x + 2$

g) $f(x) = x^2(x^2 + 3)$

$f(x) = x^4 + 3x^2$

$f'(x) = 4x^3 + 6x$

Examples: Differentiate

a) $f(x) = (1-2x)^3$

$$f(x) = -8x^3 + 12x^2 - 6x + 1$$

$$f'(x) = -24x^2 + 24x - 6$$

b) $f(x) = (x+2)(x^2-1)$

$$f(x) = x^3 + 2x^2 - x - 2$$

$$f'(x) = 3x^2 + 4x - 1$$

c) $s = \frac{3t^5 - 2t^2 + 5t^{-1}}{t^2}$

$$s = 3t^3 - 2 + 5t^{-3}$$

$$s' = 9t^2 - 15t^{-4}$$

$$s = 9t^2 - \frac{15}{t^4}$$

d) $u = \sqrt{\frac{2}{x}} + \sqrt{\frac{x}{3}}$

$$u = \sqrt{2} x^{-\frac{1}{2}} + \frac{1}{\sqrt{3}} x^{\frac{1}{2}}$$

$$u' = -\frac{\sqrt{2}}{2} x^{-\frac{3}{2}} + \frac{1}{2\sqrt{3}} x^{-\frac{1}{2}}$$

$$u' = -\frac{\sqrt{2}}{2\sqrt{x^3}} + \frac{1}{2\sqrt{3x}}$$

1. Differentiate the following functions

a) $f(x) = 3\sqrt{x} - \frac{1}{\sqrt{x}}$

$$f(x) = 3x^{\frac{1}{2}} - x^{-\frac{1}{2}}$$

$$f'(x) = \frac{3}{2\sqrt{x}} + \frac{1}{2\sqrt{x^3}}$$

b) $y = \sqrt{x}(x^3 - x^{\frac{1}{2}} + 5)$

$$y = x^{\frac{7}{2}} - x + 5x^{\frac{1}{2}}$$

$$y' = \frac{7}{2}x^{\frac{5}{2}} - 1 + \frac{5}{2}x^{-\frac{1}{2}}$$

~~$$y' = \frac{7}{2}x^{\frac{5}{2}} - 1 + \frac{5}{2}x^{-\frac{1}{2}}$$~~

c) $f(x) = \frac{2x^5 + 5x^2 + x}{3x^2}$

$$f(x) = \frac{2x^3}{3} + \frac{5}{3} + \frac{x^{-1}}{3}$$

$$f'(x) = 2x^2 - \frac{1}{3x^2}$$

$$y' = \frac{2\sqrt{x^5}}{2} - 1 + \frac{5}{2\sqrt{x}}$$

2. Find an equation of the tangent line to the curve $y = x^4 + 2x^2 - x$ at point $(1, 2)$.

$$\frac{dy}{dx} = 4x^3 + 4x - 1$$

$$\therefore y - y_1 = m(x - x_1)$$

$$\left. \frac{dy}{dx} \right|_{x=1} = 7$$

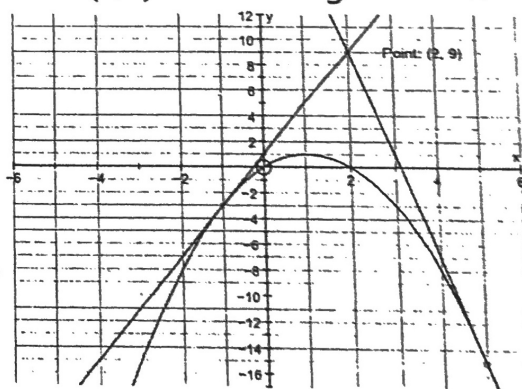
$$y - 2 = 7(x - 1)$$

3. Find the equations of both lines that pass through the point $P(2, 9)$ and are tangent to the parabola $y = 2x - x^2$. [$y = 4x + 1, y = -8x + 25$]

$$y' = -2x + 2$$

note:

$P(2, 9)$ is not on the curve.



$$y - y_1 = m(x - x_1)$$

$$2x - x^2 - 9 = (-2x + 2)(x - 2)$$

$$-x^2 + 2x - 9 = -2x^2 + 6x - 4$$

$$x^2 - 4x - 5 = 0$$

$$(x - 5)(x + 1) = 0$$

$$\therefore x = 5, -1$$

$$\textcircled{1} y - 9 = -8(x - 2)$$

$$y = -8x + 25$$

$$\textcircled{2} y - 9 = 4(x - 2)$$

$$y = 4x + 1$$

Ex. 1: Differentiate: $h(x) = \sqrt{x}(1+x)$

$$h(x) = x^{\frac{1}{2}} + x^{\frac{3}{2}}$$

$$h'(x) = \frac{1}{2} x^{-\frac{1}{2}} + \frac{3}{2} x^{\frac{1}{2}}$$

$$h'(x) = \frac{1}{2\sqrt{x}} + \frac{3\sqrt{x}}{2}$$

Ex. 2: Find the equation of the tangent line to the curve

$$f(x) = (\sqrt{x} + 2)(1 - 2\sqrt{x} + 3x) \text{ at } x = 1.$$

$$f'(x) = \frac{1 - 2\sqrt{x} + 3x}{2\sqrt{x}} + (\sqrt{x} + 2)\left(-\frac{1}{\sqrt{x}} + 3\right)$$

$$f'(1) = \frac{1 - 2 + 3}{2} + (1 + 2)(-1 + 3)$$

$$f'(1) = 1 + 6$$

$$f'(1) = 7$$

$$y - y_1 = m(x - x_1)$$

$$\therefore y - 6 = 7(x - 1)$$

Ex. 1: Differentiate $h(x) = \frac{x^2+2x-3}{x^3+1}$.

$$h'(x) = \frac{(2x+2)(x^3+1) - (x^2+2x-3)(3x^2)}{(x^3+1)^2}$$

$$h'(x) = \frac{2x^4 + 2x + 2x^3 + 2 - (3x^4 + 6x^3 - 9x^2)}{(x^3+1)^2}$$

$$h'(x) = \frac{-x^4 - 4x^3 + 9x^2 + 2x + 2}{(x^3+1)^2}$$

Ex. 2: Find $\frac{dy}{dx}$ if $y = \frac{\sqrt{x}}{1+2x}$.

$$y' = \frac{\frac{1}{2\sqrt{x}} \cdot (1+2x) - \sqrt{x} \cdot (2)}{(1+2x)^2}$$

$$y' = \frac{1+2x-4x}{2\sqrt{x}(1+2x)^2}$$

$$y' = \frac{-2x+1}{2\sqrt{x}(1+2x)^2}$$